

INFERENCES ABOUT THE COEFFICIENT OF CORRELATION IN THE STANDARD BIVARIATE NORMAL DISTRIBUTION

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Key words: Bayesian inference; bootstrap; EM algorithm; maximum likelihood; Monte Carlo; reference prior; rejection sampling.

Summary: The study of the association between two random variables that have a joint normal distribution is of interest in applied statistics; for example, in statistical genetics. This article, targeted to applied statisticians, addresses inferences about the coefficient of correlation (ρ) in the bivariate normal and standard bivariate normal distributions using likelihood, frequentist, and Bayesian perspectives. Some results are surprising. For instance, the maximum likelihood estimator and the posterior distribution of ρ in the standard bivariate normal distribution do not follow directly from results for a general bivariate normal distribution. An example employing bootstrap and rejection sampling procedures is used to illustrate some of the peculiarities.

1. Introduction

The bivariate normal distribution is ubiquitous in applications of statistics. In particular, one of its parameters, the coefficient of correlation (ρ), plays an

important role in biology. For example, the correlation between measurements taken in parents and in their offspring has an interpretation in statistical genetics in terms of “heritability”: as this correlation increases, the larger is the fraction of variance among individuals that can be attributable to genetic differences.

A special case of this distribution is the standard bivariate normal distribution, where the only unknown parameter is ρ . An application can be found in a method for describing a system of correlations, known as path analysis (Wright, 1934). Here, the inputs are the correlations, and the “path coefficients” are functions of such correlations. It appears that there is no general awareness that the maximum likelihood estimator of ρ in a standard bivariate normal distribution cannot be obtained directly from results for a general bivariate normal distribution. A similar situation occurs for the posterior distribution of ρ in a Bayesian context, as shown in this paper.

Our objective is to provide a detailed account for applied statisticians of inferences about the coefficient of correlation in a standard bivariate normal distribution both from classical and Bayesian perspectives. It is shown that the maximum likelihood estimator and the posterior distribution of ρ differ nontrivially from what it is obtained with the general bivariate normal distribution. The plan of the article is as follows. First, inferences in a bivariate normal distribution with all parameters unknown are reviewed, including incomplete data situations. Subsequently, the standard bivariate normal case is considered. A numerical example is used to illustrate developments in the last section.

2. Inferences in the bivariate normal distribution

2.1 Sampling model and classical approaches

Suppose that n vectors of order 2×1 are sampled at random, independently, from the same bivariate normal distribution. The sampling model is:

$$\begin{bmatrix} X_i \\ Y_i \end{bmatrix} \sim N \left(\begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}, \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix} \right), \quad i = 1, 2, \dots, n \quad (1)$$

and the vector of unknown parameters is $\theta = [\mu_x, \mu_y, \sigma_x, \sigma_y, \rho]'$; $-\infty < x_i, y_i, \mu_x, \mu_y < \infty$, $\sigma_x, \sigma_y > 0$ and $-1 < \rho < 1$. Here $\mu_x = E(X_i)$, $\mu_y = E(Y_i)$, $\sigma_x^2 = \text{Var}(X_i)$, $\sigma_y^2 = \text{Var}(Y_i)$ and $\rho = \text{Correlation}(X_i, Y_i)$, where $E(\cdot)$ and $\text{Var}(\cdot)$ denote expected value and variance, respectively. The probability density function (p.d.f.) of a bivariate normal vector has the form:

$$p(x, y | \theta) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \times \exp \left[-\frac{1}{2(1-\rho^2)} \left(\frac{(x-\mu_x)^2}{\sigma_x^2} + \frac{(y-\mu_y)^2}{\sigma_y^2} - 2\rho \frac{(x-\mu_x)(y-\mu_y)}{\sigma_x\sigma_y} \right) \right]. \quad (2)$$

Suppose, further, that interest is on inferring the correlation coefficient ρ from the sample of size n obtained from model (1). The joint p.d.f. of all vectors sampled is the product of n densities as in (2), and the likelihood function is the part of this joint p.d.f. that varies with θ . The maximum likelihood estimator (MLE) of ρ is known to be:

$$\hat{\rho} = \frac{\sum_{i=1}^n (X_i - \hat{\mu}_x)(Y_i - \hat{\mu}_y)}{\sqrt{\sum_{i=1}^n (X_i - \hat{\mu}_x)^2 \sum_{i=1}^n (Y_i - \hat{\mu}_y)^2}}, \quad (3)$$

where $\hat{\mu}_x = \frac{1}{n} \sum_{i=1}^n X_i$ and $\hat{\mu}_y = \frac{1}{n} \sum_{i=1}^n Y_i$ are the MLE's of μ_x and μ_y , respectively. Also $MLE(\sigma_x^2) = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu}_x)^2$ and $MLE(\sigma_y^2) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{\mu}_y)^2$. Fisher (1915) derived the sampling distribution of $\hat{\rho}$, and Stuart and Ord (1994) give as its density:

$$p(\hat{\rho} | \rho) = \frac{(1 - \rho^2)^{\frac{1}{2}(n-1)}}{n\Gamma(n-2)} (1 - \hat{\rho}^2)^{\frac{1}{2}(n-4)} \frac{d^{n-2}}{d(\hat{\rho})^{n-2}} \left\{ \frac{\arccos(-\rho\hat{\rho})}{\sqrt{1 - \rho^2\hat{\rho}^2}} \right\} \quad (4)$$

where $-1 < \hat{\rho} < 1$, and $\Gamma(\cdot)$ is the gamma function. This distribution is, evidently, difficult to work with. Fisher (1921) also found that for the random variable $Z = \log[(1 + \hat{\rho}) / (1 - \hat{\rho})] / 2$ and the parameter $\delta = \log[(1 + \rho) / (1 - \rho)] / 2$, at moderate or large n , approximately:

$$Z - \delta \sim N\left(0, (n-3)^{-1}\right). \quad (5)$$

From this, an approximate confidence interval can be developed for ρ . Also, an asymptotic confidence interval for ρ can be obtained by inverting Fisher's information matrix. This is constructed by differentiating twice the log-likelihood function with respect to θ , multiplying by -1 , and then taking expectations with respect to the sampling model (1). This does not take into account uncertainty about the other parameters (μ_x , μ_y , σ_x^2 and σ_y^2). The information matrix depends on all unknown parameters so, in practice, one must evaluate it at the MLE(θ).

An estimate of the variance of $\hat{\rho}$ or of its sampling distribution can be obtained by means of a nonparametric bootstrap procedure. Given n (x, y) pairs, a bootstrap sample consists of n pairs drawn at random and with replacement from the original pairs. Bootstrap values of $\hat{\rho}$ are calculated

for each sample, and the process can be repeated as desired. The sampling distribution of $\hat{\rho}$ can then be estimated from these replications (Efron and Tibshirani, 1993).

In practice, the data vectors may be incomplete. For example, X and Y may be measurements on pairs of twins, and X may be missing in a member of a pair while Y is observed in the co-twin. Here, one can (and often it is essential to do this) use the information from all vectors, including those with missing data, in order to improve estimation of θ . Rubin (1976) has given conditions under which a missing data process can be ignored, and the Expectation-Maximization or EM algorithm (Dempster, Laird and Rubin, 1977; McLachlan and Krishnan, 1997) is useful for maximum likelihood estimation of parameters from incomplete data. In our context, let now $n = n_x + n_y + n_{xy}$ be the total number of potentially complete vectors, n_x (n_y) be the number of vectors in which only X (Y) is observed and n_{xy} be the number of vectors in which both X and Y are observed. If the data are missing at random or if "missingness" is ignorable (Rubin, 1976; Im, Fernando and Gianola, 1989), the log-likelihood function based on the observed data has the form:

$$\begin{aligned}
 L(\theta) = & \text{constant} - \frac{n_x}{2} \log \sigma_x^2 - \frac{1}{2\sigma_x^2} \sum_{i=1}^{n_x} (x_i - \mu_x)^2 - \frac{n_y}{2} \log \sigma_y^2 \\
 & - \frac{1}{2\sigma_y^2} \sum_{i=1}^{n_y} (y_i - \mu_y)^2 - \frac{n_{xy}}{2} \log [\sigma_x^2 \sigma_y^2 (1 - \rho^2)] \\
 & - \frac{1}{2} \sum_{i=1}^{n_{xy}} [x_i - \mu_x, y_i - \mu_y] \begin{bmatrix} \sigma_x^2 & \rho \sigma_x \sigma_y \\ \rho \sigma_x \sigma_y & \sigma_y^2 \end{bmatrix}^{-1} \begin{bmatrix} x_i - \mu_x \\ y_i - \mu_y \end{bmatrix}. \quad (6)
 \end{aligned}$$

Here, the maximum likelihood estimator does not have a closed form, and a complex iterative procedure is needed to find it. However, with the EM

algorithm, one uses the form of the MLE given in (3) in the M-step, but with the missing data or functions thereof replaced by their conditional expectations, given the observed data and the current values of the parameters in the course of iteration (E-step). For example, to obtain the $(k + 1)$ th iteration for θ in the M-step, each missing value of the random variable X is replaced in the E-step by:

$$E_{\theta^{(k)}}(X_i | Y_i = y_i) = x_i^{(k)} = \mu_x^{(k)} + \frac{\sigma_{xy}^{(k)}}{\sigma_y^{2(k)}} (y_i - \mu_y^{(k)}) \quad (7)$$

and, when second-order terms are encountered, one uses:

$$E_{\theta^{(k)}}(X_i^2 | Y_i = y_i) = x_i^{(k)2} + (1 - \rho^{(k)2}) \sigma_x^{2(k)}, \quad (8)$$

where $\sigma_{xy}^{(k)} = \rho^{(k)} \sigma_x^{(k)} \sigma_y^{(k)}$ is the covariance between X and Y at iteration (k) . Similar formulae apply to missing values of Y .

When data are incomplete, the sampling distribution of $\hat{\rho}$ is unknown. Mendoza (1993) discussed the z -transformation for inferring ρ from incomplete data for two situations: missing at random and missing completely at random. With missing values, the computation of Fisher's information matrix from (6) is also more complicated than in the complete data situation, and the recovery of second derivatives in the EM algorithm requires extensive computations (Louis, 1982). The bootstrap procedure can be applied with missing data (Efron, 1994, but see cautionary comments by Rubin), although the computations are tedious, as the EM algorithm must be iterated for each bootstrap sample. Because the algorithm is slow to converge, computing time can be taxing in a sufficiently replicated bootstrap implementation. Also, unless a global maximum is located for each bootstrap sample (which is not guaranteed by the EM algorithm, Little and Rubin, 1987), the bootstrap procedure would not estimate accurately the sampling distribution of $\hat{\rho}$.

2.2 The Bayesian approach

The Bayesian view has been gaining adepts in genetics and in biostatistics (Gianola and Fernando, 1986; Berry and Stangl, 1996). Hence, it is interesting to contrast results obtained with Bayesian methods with those from classical approaches. In the Bayesian setting (Box and Tiao, 1973; Bernardo and Smith, 1994), the posterior density of all parameters, $p(\theta | data)$, is given by:

$$p(\theta | data) = \frac{p(\theta) p(data | \theta)}{p(data)} \propto p(\theta) p(data | \theta), \quad (9)$$

where $p(\theta)$ is the prior density of θ , $p(data | \theta)$ is the sampling model data density (or likelihood function, if viewed as a function of θ), and $p(data)$ is the marginal density of the data. If interest centres on any subset of θ , say θ_k , we need its marginal posterior distribution:

$$p(\theta_k | data) = \int p(\theta | data) d\theta_{-k}, \quad (10)$$

where θ_{-k} is a vector representing elements of θ not contained in θ_k , termed the “nuisance” parameters. The probability calculus (10) takes into account uncertainty about nuisance parameters automatically.

There are situations in which one does not have (or does not wish to employ) prior information about θ , so a non-informative prior must be constructed. Bernardo and Smith (1994) show that if the non-informative prior has the form:

$$p(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho) \propto \pi(\rho) \sigma_x^{-a} \sigma_y^{-a}, \quad (11)$$

where $\pi(\rho)$ is a function of ρ only, then the posterior density of ρ is:

$$p(\rho | data) = p(\rho | \hat{\rho}) \propto \pi(\rho) \frac{(1 - \rho^2)^{(n+2a-3)/2}}{(1 - \rho\hat{\rho})^{n+a-5/2}} \times F\left(\frac{1}{2}, \frac{1}{2}, n+a-\frac{3}{2}, \frac{1}{2}(1 + \rho\hat{\rho})\right), \quad (12)$$

where $F(\cdot)$ is the hypergeometric function,

$$F(a, b, c, d) = \frac{\Gamma(c)}{\Gamma(a)\Gamma(b)} \sum_{k=0}^{\infty} \frac{\Gamma(a+k)\Gamma(b+k)}{\Gamma(c+k)} \frac{d^k}{k!},$$

and $\Gamma(\cdot)$ is the gamma function. Note that the posterior distribution of ρ depends on the data only through $\hat{\rho}$.

Two non-informative prior densities of interest will be discussed here. First, we consider Jeffreys' multiparameter rule (Zellner, 1971; Box and Tiao, 1973), where the joint prior density of θ is given by:

$$p(\theta) \propto |I(\theta)|^{\frac{1}{2}}, \quad (13)$$

where

$$I(\theta) = - \int \frac{\partial^2 \log p(\text{data} | \theta)}{\partial \theta \partial \theta'} p(\text{data} | \theta) d(\text{data}) \quad (14)$$

is Fisher's information matrix. Jeffreys' rule treats all parameters as having equal importance (e.g., in our case it does not recognize ρ as being of primary interest) and does not always give satisfactory answers in the multiparameter case (Bernardo and Smith, 1994; Gelman, Carlin, Stern and Rubin, 1995). As in Sloan and Sinha (1994), we find Jeffreys' prior density to be:

$$p(\theta) \propto (1 - \ell^2)^{-2} \sigma_x^{-2} \sigma_y^{-2}. \quad (15)$$

Secondly, Bernardo (1979) suggested that a reference (prior) distribution should depend not only on the sampling model, as Jeffreys' rule does, but also on the parameter of interest. He proposed a theory for developing reference (prior) distributions based on the principle of maximizing the expected missing information (a priori). Intuitively, this would also maximize the impact of the data on inferences. This theory was applied by Bayarri (1981) to the bivariate normal distribution (1) and, for an ordered parameterization in which ρ is the

parameter of interest, she found that the reference prior density is:

$$p(\rho, \mu_x, \mu_y, \sigma_x, \sigma_y) \propto (1 - \rho^2)^{-1} \sigma_x^{-1} \sigma_y^{-1}. \quad (16)$$

Observing that this corresponds to $\pi(\rho) \propto (1 - \rho^2)^{-1}$ and to $a = 1$ in (11), the reference marginal posterior density of ρ is, from (12):

$$p(\rho | \hat{\rho}) \propto \frac{(1 - \rho^2)^{(n-3)/2}}{(1 - \rho\hat{\rho})^{n-3/2}} F\left(\frac{1}{2}, \frac{1}{2}, n - \frac{1}{2}, \frac{(1 + \rho\hat{\rho})}{2}\right). \quad (17)$$

Although the corresponding posterior distribution is not very tractable analytically, one can deal with it numerically, or draw samples of ρ by Monte Carlo methods. The distribution is guaranteed to be proper, i.e., the integral of (17) over the parameter space of ρ is finite, by construction. Also, using Monte Carlo methods, such as those based on Markov chains (e.g. Gibbs Sampling), one can accommodate missing values (for X or Y), by including these as additional unknowns in the model, and by drawing samples from their fully conditional posterior distributions in the case of the Gibbs sampler.

3. The standard bivariate normal distribution

3.1 Sampling model and classical approaches

Let now the n samples be from a standard bivariate normal distribution, so $\mu_x = \mu_y = 0$ and $\sigma_x = \sigma_y = 1$. Write:

$$\begin{bmatrix} X_i \\ Y_i \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}\right), \quad i = 1, 2, \dots, n. \quad (18)$$

Thus, each random vector follows a standard bivariate normal distribution with density:

$$p(x, y | \theta) = \frac{1}{2\pi\sqrt{1 - \rho^2}} \exp\left[-\frac{1}{2(1 - \rho^2)} (x^2 + y^2 - 2\rho xy)\right], \quad (19)$$

obtainable directly from (2) by setting the means and variances to 0 and 1, respectively. This distribution has a single parameter, so $\theta = \rho$, with $-1 < \rho < 1$, as before.

What is the maximum likelihood estimator of ρ in this model? Intuitively, there are at least three appealing candidates:

- (1) $\hat{\rho}$ as in (3), although at first examination it is not sensible to estimate μ_x and μ_y , given that these are known to be zero.
- (2) The estimator:

$$\tilde{\rho} = \frac{\sum_{i=1}^n X_i Y_i}{\sqrt{\sum_{i=1}^n X_i^2 \sum_{i=1}^n Y_i^2}}, \quad (20)$$

which would not “waste” the information used in estimating μ_x and μ_y .

- (3) A third possibility is:

$$\rho^* = \frac{1}{n} \sum_{i=1}^n X_i Y_i. \quad (21)$$

Heuristically, since it is known that $\sigma_x = \sigma_y = 1$ and that the covariance between two standardized variates is equal to its correlation, ρ^* would seem to be a logical choice. This is because in (21), knowledge that the means and variances are equal to 0 and 1, respectively, is used. It turns out, however, that none of these three estimators is the MLE of ρ in the

standard bivariate normal distribution, as shown next.

3.2 Maximum likelihood estimation

Given n independent samples drawn from a standard bivariate normal distribution, the log-likelihood function is:

$$L(\rho) = \text{constant} - \frac{n}{2} \log(1 - \rho^2) - \frac{1}{2(1 - \rho^2)} \left(\sum_{i=1}^n (x_i^2 + y_i^2) - 2\rho \sum_{i=1}^n x_i y_i \right). \quad (22)$$

Its first derivative with respect to ρ is:

$$\begin{aligned} \frac{\partial L(\rho)}{\partial \rho} &= \frac{n\rho}{1 - \rho^2} - \frac{\rho}{(1 - \rho^2)^2} \left(\sum_{i=1}^n (x_i^2 + y_i^2) - 2\rho \sum_{i=1}^n x_i y_i \right) \\ &\quad + \frac{1}{1 - \rho^2} \sum_{i=1}^n x_i y_i. \end{aligned} \quad (23)$$

To locate a maximum, (23) must be set to zero. After rearrangement, the ML estimate of ρ must be one of the solutions to the cubic equation:

$$n\hat{\rho}^3 - a\hat{\rho}^2 + (b - n)\hat{\rho} - a = 0, \quad (24)$$

where $a = \sum_{i=1}^n x_i y_i$ and $b = \sum_{i=1}^n (x_i^2 + y_i^2)$. Letting $a' = a/n$ and $b' = (b/n - 1)$, the solutions to (24) are:

$$\hat{\rho}_1 = \frac{1}{3} (a' - c_1 + c_2); \quad \hat{\rho}_2 = \frac{1}{3} \left(a' + \left(\frac{1 + 3I}{2} \right) c_1 - \left(\frac{1 - 3I}{2} \right) c_2 \right)$$

and

$$\hat{\rho}_3 = \frac{1}{3} \left(a' + \left(\frac{1-3I}{2} \right) c_1 - \left(\frac{1+3I}{2} \right) c_2 \right),$$

where $I = \sqrt{-1}$; $c_1 = \frac{\sqrt[3]{2}(3b' - a'^2)}{k}$; $c_2 = \frac{k}{(3\sqrt[3]{2})}$; and

$$k = \sqrt[3]{27a' + 2a'^3 - 9a'b' + \sqrt{27(27a'^2 + 4a'^4 - 18a'^2b' - a'^2b'^2 + 4b'^3)}}$$

Two of the three roots may be complex, but there is always at least one real root in the space $-1 < \rho < 1$ (Stuart and Ord, 1994). This is so because the function:

$$g(\rho) = n\rho^3 - a\rho^2 + (b - n)\rho - a$$

is such that

$$g(-1) = -\sum_{i=1}^n (x_i + y_i)^2 < 0$$

and

$$g(1) = \sum_{i=1}^n (x_i - y_i)^2 > 0.$$

If more than one real root lies inside this interval, we take as MLE the one producing the largest value of the likelihood function. In Figure 1 an example is given where 15 vectors were drawn at random from a standard bivariate normal distribution with $\rho = 0.8$. Here, the ML equation has three real roots and the ML estimate of ρ is $\hat{\rho} = 0.814$. The data set is in the Appendix (see situation 1: "Three real roots").

As $n \rightarrow \infty$, there will be just one real root to the maximum likelihood equation. Dividing (24) by n and taking limits, the left-hand side becomes:

$$\begin{aligned}
 & \lim_{n \rightarrow \infty} \left[\hat{\rho}^3 - \frac{a}{n} \hat{\rho}^2 + \left(\frac{b}{n} - 1 \right) \hat{\rho} - \frac{a}{n} \right] \\
 &= \hat{\rho}^3 - \lim_{n \rightarrow \infty} \left[\frac{a}{n} \right] \hat{\rho}^2 + \left(\lim_{n \rightarrow \infty} \left[\frac{b}{n} \right] - 1 \right) \hat{\rho} - \lim_{n \rightarrow \infty} \left[\frac{a}{n} \right] \\
 &= \hat{\rho}^3 - E[XY] \hat{\rho}^2 + (E[X^2 + Y^2] - 1) \hat{\rho} - E[XY] \\
 &= \hat{\rho}^3 - \rho \hat{\rho}^2 + \hat{\rho} - \rho.
 \end{aligned} \tag{25}$$

Rearranging and equating (25) to zero we get $(\hat{\rho}^2 + 1)(\hat{\rho} - \rho) = 0$, which has the unique real solution $\hat{\rho} = \rho$, confirming the consistency property of the MLE.

An interesting situation arises when the sum of products $a = 0$. In this case, equation (24) reduces to $n\hat{\rho}^3 + (b - n)\hat{\rho} = 0$ and then: 1) if $b \geq n$, there will be just one real root $\hat{\rho} = 0$, which will be the maximum likelihood estimate. 2) if $b < n$, the roots will be $\hat{\rho}_1 = 0$, associated with a local minimum, whereas $\hat{\rho}_2 = +\sqrt{1 - b/n}$ and $\hat{\rho}_3 = -\sqrt{1 - b/n}$, will have opposite signs but same likelihood value! These two situations are illustrated in Figure 2. In the first column, two different samples of four data points, each having $b = 8 > n = 4$ are plotted in (a1) and (b1). The likelihood is identical for both sets, as shown in (c1), and the maximum likelihood estimator is 0. In the second column, two data sets with $b = 2 < n = 4$ are plotted in (a2) and (b2). The likelihood and its three stationary points are in (c2). Just a

difference in scale between examples on the same row of Figure 2 completely changes the results. It is seen that in the standard bivariate normal distribution $\sum_{i=1}^n x_i y_i = 0$ does not imply that the maximum likelihood estimate of ρ is 0.

3.3 Missing data in the standard bivariate normal distribution

When there are missing data, the log-likelihood function for the standard bivariate normal distribution is obtained by putting $\mu_x = \mu_y = 0$ and $\sigma_x = \sigma_y = 1$ in (6). Then, all terms other than the last two ones become constants so:

$$L(\rho) = \text{constant} - \frac{n_{xy}}{2} \log(1 - \rho^2) - \frac{1}{2} \sum_{i=1}^{n_{xy}} [x_i, y_i] \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}^{-1} \begin{bmatrix} x_i \\ y_i \end{bmatrix}. \quad (26)$$

In contrast to the situation in the bivariate normal distribution, missing data do not provide any information about ρ . It is instructive to note that the EM algorithm recognizes this, as illustrated by the following example. Suppose there are n complete pairs and that there is an additional pair where only $X = x_{n+1}$ is observed. The E-step requires the imputations:

$$E_{\hat{\rho}}(Y_{n+1} | X_{n+1} = x_{n+1}) = \hat{\rho} x_{n+1} \text{ and}$$

$$E_{\hat{\rho}}(Y_{n+1}^2 | X_{n+1} = x_{n+1}) = 1 + \hat{\rho}^2 (x_{n+1}^2 - 1)$$

where $\hat{\rho}$ is the current value of ρ , and the M-step requires solving (24) but now with $n + 1$ pairs. The cubic equation is then:

$$(n + 1) \hat{\rho}^3 - a^* \hat{\rho}^2 + (b^* - (n + 1)) \hat{\rho} - a^* = 0. \quad (27)$$

Here

$$a^* = a + x_{n+1} E_{\hat{\rho}} (Y_{n+1} | X_{n+1} = x_{n+1}) = \sum_{i=1}^n x_i y_i + \hat{\rho} x_{n+1}^2$$

and:

$$\begin{aligned} b^* &= b + x_{n+1}^2 + E_{\hat{\rho}} (Y_{n+1}^2 | X_{n+1} = x_{n+1}) \\ &= \sum_{i=1}^{n+1} x_i^2 + \sum_{i=1}^n y_i^2 + (1 + \hat{\rho}^2 (x_{n+1}^2 - 1)) , \end{aligned}$$

with a and b as before. It can be verified that the left-hand side of (27) is precisely (24), thus leading to the same solution, $\hat{\rho}$, obtained by using the complete pairs only.

3.4 Bayesian approach with Jeffreys' prior

Jeffreys' prior for the bivariate normal distribution is in (15), and the ρ -reference prior is in (16). Because in the standard bivariate normal distribution the mean and the variance are known, one could think naively that the density of the non-informative prior distribution of ρ is either $p(\rho) \propto (1 - \rho^2)^{-2}$ or $p(\rho) \propto (1 - \rho^2)^{-1}$. It turns out that none of these is the non-informative prior density sought. Applying Jeffreys' rule, we find that:

$$p(\rho) \propto \frac{\sqrt{1 + \rho^2}}{(1 - \rho^2)} . \quad (28)$$

This is also the reference prior because in the single parameter case, Jeffreys' rule gives directly the reference prior (Bernardo, 1979). The posterior distribution of ρ is thus expressible as:

$$p(\rho | \text{data}) \propto \frac{\sqrt{1 + \rho^2}}{(1 - \rho^2)^{(n/2+1)}}$$

$$\times \exp \left[-\frac{1}{2(1-\rho^2)} \left(\sum_{i=1}^n (x_i^2 + y_i^2) - 2\rho \sum_{i=1}^n x_i y_i \right) \right]. \quad (29)$$

This does not have a recognizable form, and cannot be put in the “canonical” form (12) because the maximum likelihood estimator of ρ cannot be written explicitly, contrary to what occurs in the bivariate normal distribution.

4. Illustration of likelihood and Bayesian inference

Data were simulated to illustrate the procedures discussed in Sections 2 and 3.

Twenty thousand independent random samples consisting of 15 pairs each were drawn from a standard bivariate normal distribution with $\rho = 0.8$. Histogram (a1) in Figure 3 gives the distribution of estimates of ρ obtained with (3) and histogram (a2) depicts the distribution of the MLE found solving (24).

The maximum likelihood estimator calculated from a general bivariate normal distribution is less precise, which is evident from comparison of histograms (a1) and (a2). Table 1 gives some summary statistics for this and other situations.

An analysis of a single sample with 15 pairs (see “Analysis” data set in the Appendix and in Figure 4) drawn from the same standard bivariate normal distribution, with $\rho = 0.8$, was carried out. Histograms (b1) and (b2) in Figure 3 give the bootstrap distribution of the estimates found from 20,000 bootstrap replicates from the “Analysis” sample. The bootstrap histograms agreed well with the empirical distributions in (a1) and (a2), although the range between the 2.5 and 97.5 percentiles was narrower. The superiority of the “correct” estimator holds in this single sample analysis as confirmed by the statistics in Table 1.

A Bayesian analysis of the "Analysis" data set was undertaken as well. Assuming (incorrectly) a bivariate normal sampling model, 20,000 samples were drawn from the reference marginal posterior distribution (17). Also, 20,000 samples were drawn from the reference posterior for the standard bivariate normal model in (29). A rejection sampling scheme (Ripley, 1987) was adopted to draw from the appropriate posterior distribution of ρ . Let $f(\rho) = \text{uniform}(-1, 1)$ density and $m = \max[p(\rho | \text{data})]$, where $p(\rho | \text{data})$ is either (17) or (29). The following steps were used to obtain the random draw of ρ , $\rho_{(k)}$, $k = 1, 2, \dots, 20,000$, from $p(\rho | \text{data})$:

1. Generate a random deviate ρ^* from $f(\rho)$;
2. Generate a $\text{uniform}(0, 1)$ variate u ;
3. If $\frac{p(\rho^* | \text{data})}{m} > u$, $\rho^{(k)} = \rho^*$, otherwise return to step 1.

The hyper-geometric function $F(\cdot)$ was approximated by stopping the series when any individual term was smaller than 10^{-17} . The estimated posterior distributions are shown in histograms (c1) and (c2) of Figure 3. The reference posterior distributions were slightly more skewed than the corresponding sampling or bootstrap distributions and had longer left tails. The 95% credibility sets (Table 1) had roughly the same width in the two posterior distributions, in contrast with the sampling distribution, where the empirical inter-percentiles interval was narrower when the standard bivariate normal sampling model was employed in the analysis.

5. Conclusion

Contrary to what would be expected intuitively, the maximum likelihood analysis of a standard bivariate normal distribution is conceptually and numerically more complex than that of a bivariate normal distribution with all five parameters unknown. If the sampling model is truly that of the standard bivariate normal distribution, sharper inferences about the correlation are obtained by using the appropriate maximum likelihood estimator. It is relatively straightforward to carry out a reference Bayesian analysis of the correlation coefficient for any of the two sampling models.

Acknowledgements

This work was supported by the United States Department of Agriculture (Hatch Project 3733), by research grant NRICGP/USDA 99-35205-8162, and by a grant from the Fundação de Amparo a Pesquisa do Estado de São Paulo (FAPESP), Brazil. The authors wish to thank M.J. Bayarri, J.M. Bernardo and M.A. Newton for useful comments.

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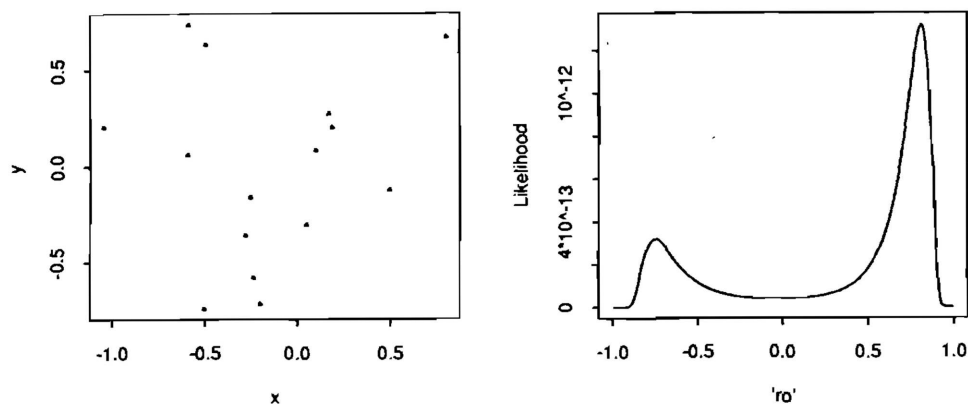
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Table 1. Mean, median, standard deviation (SD) and percentiles for each of the distributions in Figure 3.

	Histogram					
	(a1)	(a2)	(b1)	(b2)	(c1)	(c2)
Mean	0.789	0.800	0.815	0.802	0.810	0.771
Median	0.811	0.809	0.829	0.808	0.830	0.792
SD	0.109	0.073	0.087	0.079	0.100	0.096
$L^{(1)}$	0.519	0.632	0.608	0.631	0.556	0.528
$U^{(2)}$	0.934	0.916	0.946	0.933	0.941	0.893

(1), (2) 2.5 and 97.5 percentiles, respectively, of the appropriate distribution.
For the posterior distributions (c1) and (c2), this gives a 95% credibility set.

**Figure 1.** Scatter diagram and likelihood function for a randomly drawn data set ($n = 15$) from a standard bivariate normal distribution ($\rho = 0.8$). Here the ML equation has three real roots

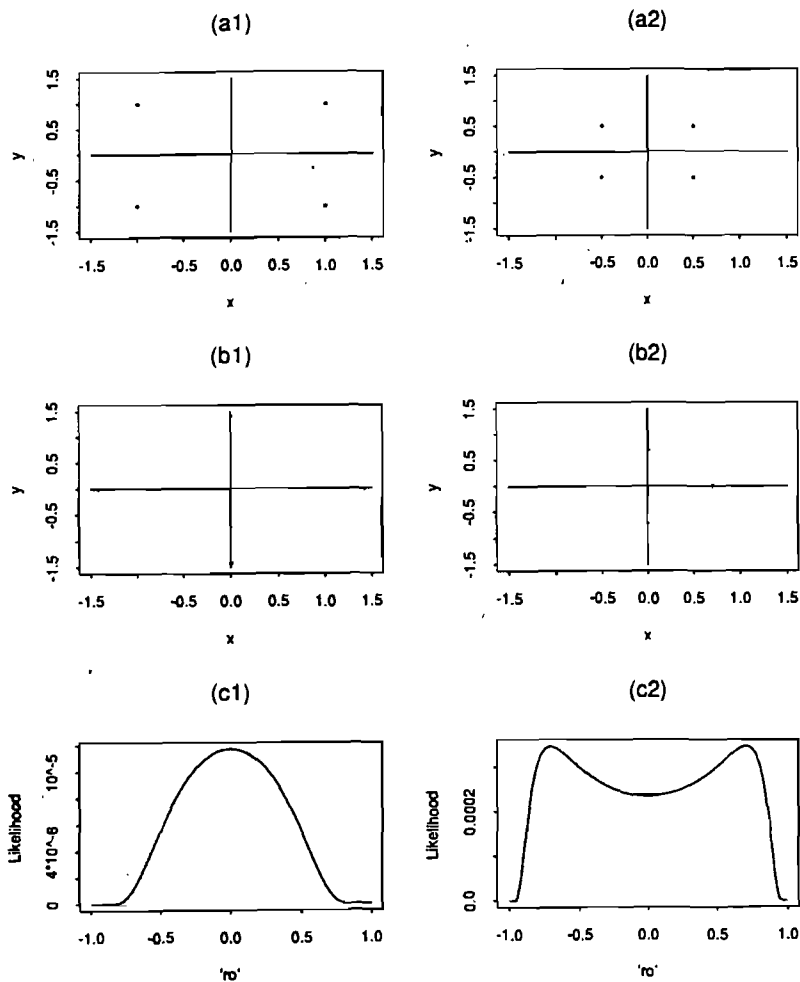


Figure 2. Scatter diagrams for four example data sets where $\sum xy = 0$ and $n = 4$, and the respective likelihood functions of ρ . In the first column $\sum (x^2 + y^2) = 8$ in the data sets (a1) and (b1). In the second column $\sum (x^2 + y^2) = 2$ for the data sets (a2) and (b2).

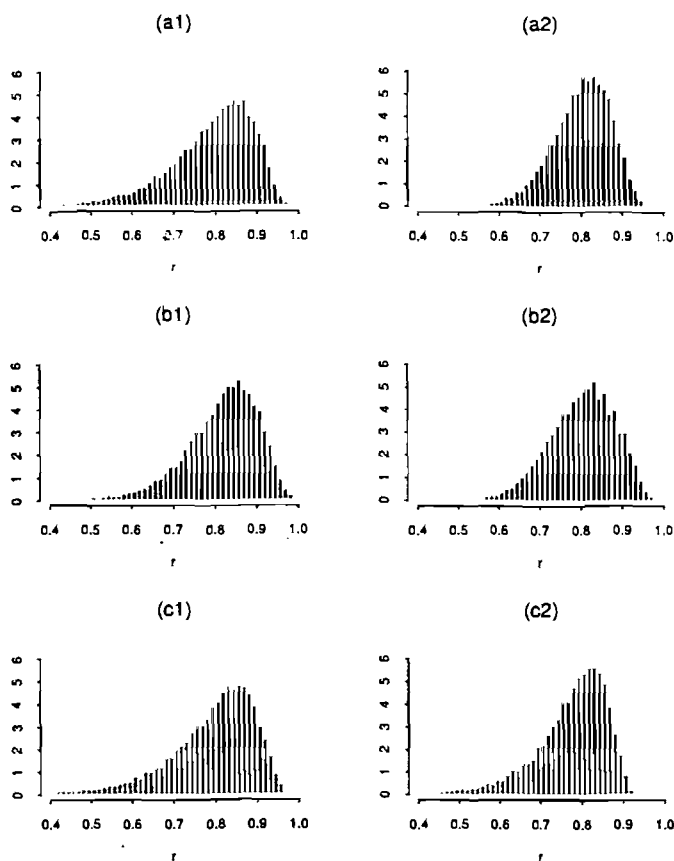


Figure 3. Histograms of sampling and bootstrap distributions of maximum likelihood estimates of the correlation coefficient ρ and of its reference posterior distributions. (a1) and (a2) are distributions of 20,000 estimates of ρ in the bivariate normal and standard bivariate normal distributions (SBVN), respectively, calculated from the same samples of size 15 drawn from a SBVN with $\rho = 0.8$. (b1) and (b2) are the corresponding bootstrap distributions for a single sample ("Analysis" in the Appendix). (c1) and (c2) are the corresponding reference posterior distributions for the "Analysis" sample.

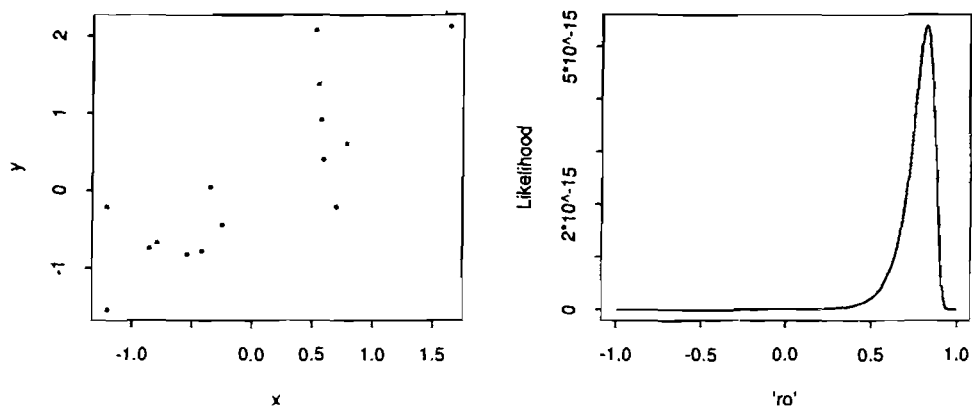


Figure 4. Random data set ($n = 15$) from a standard bivariate normal distribution with correlation coefficient $\rho = 0.8$ (see the Appendix), and likelihood function of ρ .

Appendix: Example data sets: (1) “Three real roots”. (2) “Analysis”.

pair (i)	Situation			
	(1) “Three real roots”		(2) “Analysis”	
	x_i	y_i	x_i	y_i
1	-1.0459	0.2051	0.7795	0.6029
2	0.0971	0.0817	-0.3464	0.0469
3	-0.2015	-0.7167	1.6410	2.1270
4	-0.5898	0.7290	0.5730	0.9080
5	0.0451	-0.3020	-0.7915	-0.6653
6	-0.5911	0.0559	-0.4195	-0.7704
7	-0.2559	-0.1659	0.5484	1.3760
8	-0.2813	-0.3610	-0.2444	-0.4453
9	0.7971	0.6633	-1.2185	-0.2124
10	0.1874	0.1961	0.5207	2.073
11	-0.2368	-0.5812	-0.8567	-0.7396
12	0.1664	0.2661	0.6963	-0.2153
13	-0.4961	0.6252	-0.5427	-0.8255
14	-0.5021	-0.7388	0.5881	0.3942
15	0.4884	-0.1230	-1.2120	-1.5300